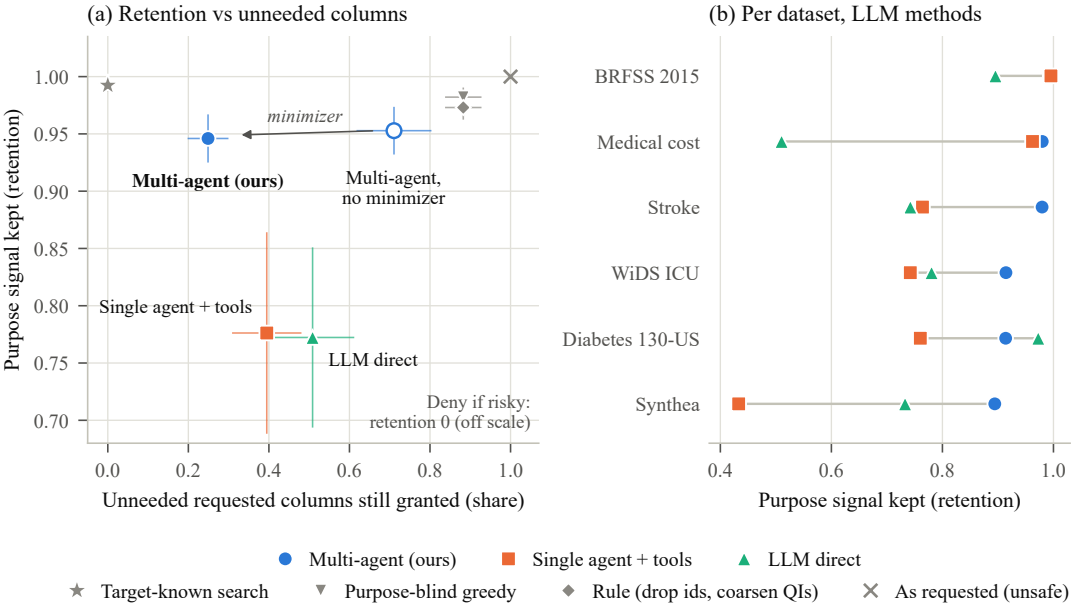

(a) Retention vs unneeded columns
Purpose signal kept (retention)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
minimizer
Multi-agent (ours)
Multi-agent,
no minimizer
Single agent + tools
LLM direct
Deny if risky:
retention 0 (off scale)
0.0
0.2
0.4
0.6
0.8
1.0
Unneeded requested columns still granted (share)
(b) Per dataset, LLM methods
BRFSS 2015
Medical cost
Stroke
WiDS ICU
Diabetes 130-US
Synthea
0.4
0.6
0.8
1.0
Purpose signal kept (retention)
Multi-agent (ours)
Single agent + tools
LLM direct
Target-known search
Purpose-blind greedy
Rule (drop ids, coarsen QIs)
As requested (unsafe)